

Chapter 3

LINUX System and Network Administration

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2025/2026



Outlines

- **Introduction**
- **Linux Systems and Network Concepts**
- **Linux User Administration**
- **Linux Service/Server Administration**
 - **Supporting a Windows Network - through SAMBA**
 - **Mail Server**

Unix Operating system

- It is a **multi-user**, **multi-tasking** operating system.
- That means, many users logged into a system simultaneously, each running many programs
- It's the **kernel's** job to **keep each process** and **user separate** and to regulate access to system hardware, including CPU, memory, disk and other I/O devices

Linux Operating System

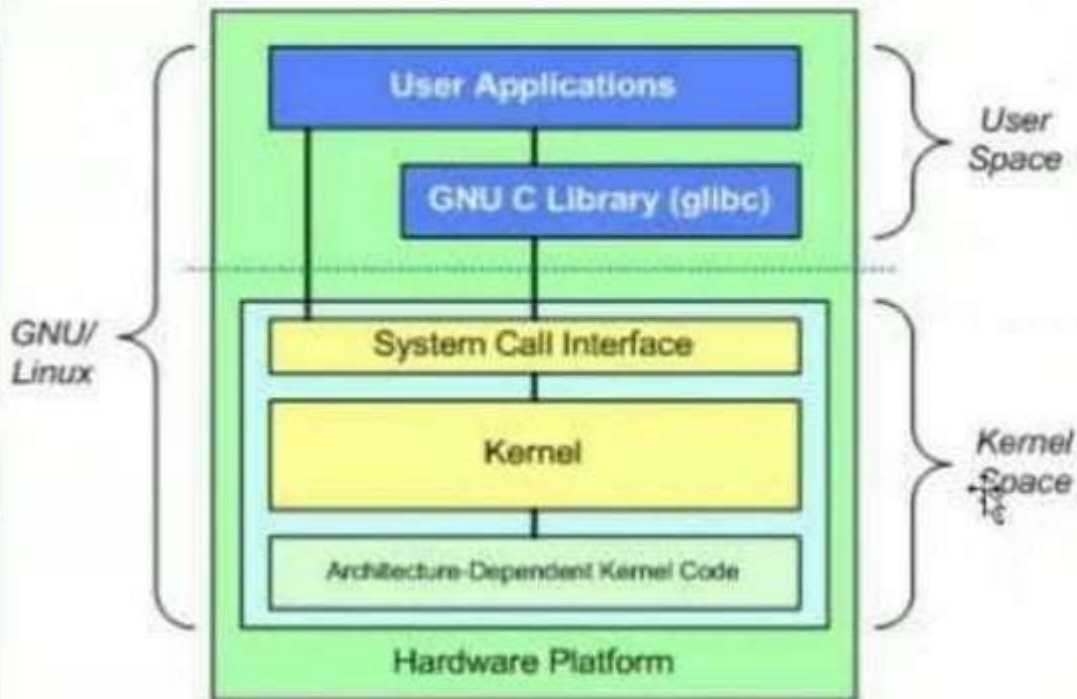
• What is Linux?

- It is a **free** and **open** source Unix-type operating system
- It provides **programming interface** as well as **user interface** compatible with Unix
- Developed under GNU General Public License, the source code for Linux is **freely available to everyone**
- Linux was first released in the year **1991**.
- **Debian**, **Fedora**, and **Ubuntu** are the **popular Linux distributions**. Which one we have in the lab installed?
- **RedHat Enterprise Linux** and **SUSE Linux Enterprise Server** (SLES) that are available as **commercial distributions** of Linuxss

Linux Architecture

- As it is freely redistributable, anyone can modify and create variations of source code.
- Two layers of Linux are **User space** and **Kernel space**.
- Within these layers, there are **four** main components i.e. **Hardware, Kernel, System call interface (aka Shell)** and **User applications**
- **Hardware** is comprised of all the peripheral devices which are attached to the computer such as **terminals, printer, CPU, RAM, mother board**

Linux Architecture...



Linux Architecture....

- **Kernel:** It is responsible for a lot of critical tasks like:
 - Process control,
 - Networking management,
 - Accessing peripherals and file system,
 - Security management and
 - Memory management.

Distinction between Linux and Windows OS

Linux OS	Windows OS
➤ The Linux kernel released on 17th September, 1991	➤ Windows was first released in 1985.
➤ Linux was developed by Linus Torvalds	➤ Windows OS was developed by Microsoft
➤ The first version of Linux was Version 0.01	➤ The first version of Windows OS was Windows 1.0
➤ Linux combined with the GNU system to produce a complete free operating system	➤ Windows OS is neither open source nor free OS and is the property of Microsoft Company

Linux is completely freed from price whereas windows is marketable package and is expensive

Linux Systems and Network Concepts

- **Linux systems** support a wide range of network concepts and technologies, making a **popular choice for network**
- **IP Addressing**
 - Support both IPv4 and IPv6 addressing used to identify devices and communicate over a network
- **Network Interfaces:** physical or virtual devices used to connect a Linux system to a network
- **Routing:** is the process of forwarding data between networks.
- **Firewalls:** used to protect Linux systems from unauthorized access and can be configured to filter incoming and outgoing network traffic

Linux Systems and Network Concepts

- Seven OSI layers
- Four DoD layers: the four DoD (Department of Defense) layers, also known as the TCP/IP model,
- **Wireshark sniffer**: network protocol analyzer and sniffer tool that is used to **monitor and analyze network traffic**
- **Sniffing for ntp** (Network Time Protocol) packets
- **Sniffing an ARP** (Address Resolution Protocol) broadcast can capture and display information about the ARP requests and responses, including the MAC addresses of the devices on the network and corresponding IP addresses

Linux Systems and Network Concepts ...

- **Unicast**: originates from one computer and is destined for exactly one other computer
- **Multicast** is destined for a group of computers
- **A broadcast** is meant for everyone.
- **An anycast** signal goes to the geographically nearest of a well defined group

Linux File System

- *The Linux file system organizes data on a storage device into **files** and **directories**.*
- **Files** can contain data or programs
- **Directories** *organize files into a hierarchical structure.*

File System Hierarchy

- **Linux file system hierarchy** is organized like a **tree**, with the root directory at the top and sub-directories branching out from there.
- Some of the main directories in the Linux file system include:
 - **/:** The root directory of the file system.
 - **/bin:** Contains essential user binaries, such as **ls** and **cp** commands.

Linux File System ...

File System Hierarchy ...

- **/sbin**: Contains essential **system binaries**, such as those used to manage the system network.
- **/etc**: Contains system **configuration files**, such as those used to **configure the network or set up user accounts**.
- **/usr**: Contains **user programs and files**, such as documentation and libraries.
- **/var**: Contains **variable data files**, such as **log files and mail spools**
- **/tmp**: Contains **temporary files** that are deleted when the system is rebooted.
- **/home**: Contains **user home directories**

Linux File System ...

File System Types

Linux supports several file system types, including:

- **ext4**: the default file system for most **modern** Linux distributions.
- **XFS**: a high-performance file system designed for large-scale environment
- **btrfs**: *a modern file system that supports features such as snapshots and compression.*
- **NTFS**: *a file system used by Windows that can be mounted on Linux systems.*

Linux File System ...

Mounting File System

- To access data on a storage device, the file system must be mounted
- This involves attaching the file system to a directory in the existing file system hierarchy
- The **mount** command is used to mount a file system, and **umount** command is used to unmount it.

Permissions and Ownership

- *Linux uses a **permission system** to control access to **files and directories**.*
- *Each **file and directory** has an **owner** and a **group**, and **three** sets of permissions: **read**, **write**, and **execute**.*
- *These permissions can be set for the **owner**, the **group**, and **all other users***

Permissions and Groups

- Permissions are divided into **three user-based groups**:
 1. **Owner**: apply only to the user who owns the file or directory
 - Owner can **read**, **write**, and **execute** the file or directory; and **can also change its permissions**.
 2. **Group**: apply only to the group assigned to the file or directory
 - Group can read, write, and execute the file or directory but other users who are not part of the group cannot
 3. **All users**: apply to all other users on the system who are not owner or part of the group assigned to the file or directory
 - These users can read, write, and execute the file or directory **if the other permissions allow it**

Permission Types

- There are **three** types of permissions in Linux
- **Read (r)**: Allows a user to **read** the contents of a file or **list** the contents of a directory.
- **Write (w)**: Allows a user to **modify** the contents of a file or **create**, *rename* or **delete** files in a directory:
- **Execute (x)**. Allows a user to **execute a file** or **enter a directory**
- **No permission (-)**: user may not have any permission for reading, writing or execution

Permission Types ...

- In general, access permissions specify who can access a particular file or resource and what actions they can perform on it such as reading, writing or executing
- To change permissions in Linux, we can use **chmod** command
- syntax for the **chmod** command:
- **chmod [permissions] [file/directory]**
- or
- **chmod [user] [operation][permissions] [file/directory]**

Permission Types ...

Example 1:

- *If you are the owner of a file called **account** and want to change the permissions or modes so that:*
 - **User** can *read, write and execute*,
 - **Group** members can *read and execute only* and
 - **Others** can only *read*, you will run the commands bellow:

sudo chmod u=rwx, g=rx, o=r account

endetx@Ubuntu:~/Desktop\$ *chmod u=rwx, g=rx, o=r account*

endetx@Ubuntu:~/Desktop\$ *ls -l*

drwxr-xr-- endex 4096 Jan 22 06:54 account

d = directory

Permission Types ...

Examples 2:

- *On the other hand if you permit read, write and execute for type of user permissions for a file **account***
- **User**, **group** and **other users** can **rend**, **write** and **execute** the **chmod** will be as follows:

sudo chmod u=rwx, g=rwx, o=rwx account

Exercise 1:

*If you change permission mode for a file **exam** as follows what **chmod** command you used? For **user** read and write only for **group** read only and for **other** write and execute; what **chmod** command used?*

Answer: ***sudo chmod u=rw, g=r, o=wx exam***

Permission Types ...

About **chmod** command:

- The **chmod** command changes the permission of the other file called confidential with permission that user can read (r), write (w) and execute (x) it

Group members can only read (r) and execute (x) and
Others can only read (r) its content...

The same command above can be written as shown below
using its **octal** permission notation

chmod 754 confidential

Modifications of Permission

About **chmod** command:

The same command on previous slide can be written as shown below using its **octal** permission notation.

```
sudo chmod 754 confidential
```

The digit 7 is equal to **read**, **write** and **execute**. The digit 5 is equal to **read** and **execute**, the digit 4 is equal **read**.

7 = read, write, execute

5 = read, execute

*4 = **Read***

*2 = **Write***

*1 = **Execute***

0 = no permission

So, $7 = 4 + 2 + 1$ (rwx);

$5 = 4 + 1$ (rx);

4 (r)

Viewing the Permissions

permissions view by checking the file or directory permissions by using

`ls -l` command.

```
endex@ubuntu:~$ ls
Desktop  Downloads  Music  Public  Videos
Documents  examples.desktop  Pictures  Templates
```

```
endex@ubuntu:~$ ls -l
total 44
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Desktop
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Documents
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Downloads
-rw-r--r-- 1 endex endex 8980 Dec 26 2019 examples.desktop
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Music
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Pictures
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Public
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Templates
drwxr-xr-x 2 endex endex 4096 Dec 26 04:24 Videos
endex@ubuntu:~$
```


Linux Commands for System Administrators

- *Commands are used in Linux for different advantages:*
 - **It can save time**
 - **Used when unable to use GUI**
- **Some commands used in Linux:**
 - **ls** Used to *list* files and folders in the current working directories
 - **ls -l** Used to *Sort* file as per last modified *time* and *permission* type for file or directory
 - **cd** Used to *change directory* from the current you are working on to other
 - **cd/** Take you to the root directory

Example: **root/d1/d2/d4\$ cd ...**

Becomes **root\$**

Linux Commands for System Administrators ...

Some commands used in Linux: ...

- **cd ..** Used to **return back** one directory from the current directory

Example: **root/d1/d2/d4\$ cd ..** Becomes
root/d1/d2>

- **apt -get** Used to **install**, **upgrade** and **update** software

Example: **root/d1/d2/d4\$ apt —get install sw1**

- **pwd** Used to **print the full path** of current working directory (print working directory)

Example: **admin/Desktop/d2/d4\$ pwd** will display
./home/admin/Desktop/d2/d4\$

Linux Commands for System Administrators ...

Some common commands used in Linux...

- **cp** Used to **copy file**. You should to specify both a **file** you copy and **location** where you going to copy

Example: **cp** /home/admin/Desktop/**sata**\$ file.odt

/home/admin/Desktop/**new**\$

- This example copies a file called file.odt from *sata* directory to *new* directory

Linux Commands for System Administrators ...

Some common commands used in Linux ...

mkdir/ Used to create new directories (also known as folders) in the current directory)

Exemple: `/admin/Desktop$ mkdir folder1`

rmdir/ Used to remove or delete directory from the system

Example: `/admin/Desktop$ rmdir folder1`

This syntax is used to remove folder **folder1** from Desktop

rm Used to remove files from folder

Example: `/admin/Desktop/folder1$ rm file1`

This syntax is used to remove a file called **file1** from folder called **folder1**

Linux Commands for System Administrators ...

Some common commands used in Linux ...

- **W** *Used to display users currently **logged** in and their process along with shows **load averages***
 - *Also shows the **login name**, **tty name**, **remote host**, **login time**, **idle time**.*
 - *An options that used with this commands:*
 - h : displays no header entries.**
 - s : without JCPU and PCPU.**
 - f: Removes from field.**
 - V : (upper letts) Shows versions**

Linux Commands for System Administrators ...

- **users** *Used to display currently logged in users*
 - *This command don't have other parameters other than help and version.*
- **who** *Used to display **user name**, **date**, **time** and host information.*
 - *These are an options for this command*
 - b : Displays last system reboot date and time.**
 - r : Shows current runlet.**
 - a, - all: Displays all information in cumulatively.**
- **mv** *Used to rename file or folder*
 - **Example:** **mv** *old new* (to rename folder *old* to *new*)

Linux Commands for System Administrators ...

- **mv** Also used to *move files or folder* to other

Example: `/home/endex/Desktop/old$ mv file1.odt
/home/endex/Desktop$ new`

This is *file1.odt* is moved from current folder to folder called **new**

Less It allows *quickly to view file*.

Also can page *up* and *down*.

Press '*q*' to quit from less window.

For more commands (Please, **read**)

Linux Commands and Tools for System Administrators

Some common **Network** commands used in Linux

- **Vmstat**
- **Tcpdump**
- **Netstat**
- **Ping**
- **Route**
- **Traceroute**
- **Ifconfig**
- **Nslookup**
- **Hostname**

Linux Commands and Tools for System Administrators...

- **IoStat: Input / Output Statistics**
 - It is a tool used to collect and show system **input and output storage device statistics**
 - This tool is often used to **trace storage device performance**
 - But it should be **installed** first to use it

Linux Commands and Tools for System Administrators...

VmStat: Virtual Memory Statistics

This command used to display Statistics of the following:

- virtual memory,
 - kernel threads,
 - interrupts,
 - *disks*,
 - *System processes*
 - *I/O blocks*,
 - CPU activity and much more.
- By default **vmstat** command is not available under Linux systems you need to **install** a package called **sysstat**

```
endex@ubuntu:~$ vmstat
procs -----memory----- ---swap-- -----io---- -system-- -----cpu-----
 r b  swpd  free  buff  cache   si   so    bi   bo    in   cs us sy id wa st
  1  0      0 22352 46624 528488    0    0   564   14  102  434  6  6 86  3  0
endex@ubuntu:~$ |
```

Linux Commands and Tools for System Administrators...

- **Lsof** : List Open Files
 - Used to display list of the *open files* and the *processes*
 - The **open files** included are:
 - Disk files,
 - Network sockets,
 - List of devices and processes.
- **netstat** : Network Statistics
 - Used to display *network connection, routing tables, and network interface statistics*
- **Tcpdump**: allows you to *capture and analyze network traffic in real-time*

Linux Commands and Tools for System Administrators...

- **ifconfig** (interface configurator):
 - To initialize an interface,
 - *To assign IP Address to interface and enable or*
 - *To disable interface on demand.*
 - *To view IP Address and Hardware / MAC Address*
 - *To view MTU (Maximum transmission unit) size*

Example to assign IP address for interface *eth0*

```
/home/endex/$ ifconfig eth0 192.168.50.5 netmask 255.255.255.0
```

```
/home/endex /$ ifconfig eth0 (initialize interface)
```

```
/home/endex/$ ifup eth0 (enable eth0 interface)
```

```
/home/endex/$ ifdown eth0 (disable eth0 interface)
```

Linux Commands and Tools for System Administrators...

- **traceroute** *Used to trace the path that packets take from the local system to a remote host*

Example: *below let tracing route to global DNS*

- *endex/ubuntu\$ traceroute 4.2.2.2*

- **nslookup** : *use to find out DNS related query*

The following examples shows A Record (IP Address of www.google.com)

Example: **endex/ubuntu\$ www.google.com**

Linux Commands and Tools for System Administrators...

- **route**: is used to display and modify the system's network **routing table**

Example *endex/ubuntu\$ route*

- **ARP** is useful to **view / add** the contents of the kernel's **ARP tables**

Example *to see default table use the command as*

endex/ubuntu\$ arp -e

- **Ping** (*Packet INternet Groper*)

*Used to **test network connectivity** to a remote host by sending packets to the host and waiting for response.*

Example: *erndex/ubutitu\$ ping 4.2.2.2 OR*

endex/ubuntu\$ www.google.com

Linux Commands and Tools for System Administrators...

• Host

- Used to **find name to IP** or **IP to name** in IPv4 or IPv6 and
- .Example: endex/ubuntu\$ **host** www.google.com **OR**
endex/ubuntu\$ **www.google.com**

• iwconfig command

- use to **configure a wireless network** interface.
- set the basic Wi-Fi details like: SSID channel and encryption.
- endex/ubuntu\$ **iwconfig [interface]**

Linux User Administration

- Involves managing user accounts and their associated permissions.
- Common tasks involved in Linux user administration:
 - **Creating User Accounts:** used **useradd** [options] username
 - **Managing User Groups:** Users can be organized into groups to simplify permissions management.
 - **To create a new group, you can use **groupadd** command.**
 - **groupadd** *groupname*
 - *To add a user to a group, you can use the **usermod** command.*
 - **usermod** *-aG groupname username*

Linux User Administration ...

- **Setting User Permissions:** to set permissions for a file or directory, you can use the **chmod** command
 - **chmod [permissions] filename**
- **Setting User Quotas:** you can set **disk quotas** to limit the amount of disk space that each user can use.
- To set a quota for a user, you can use the **edquota** command
 - **edquota -u username**
- **Removing User Accounts:** you can use the **userdel** command
 - **userdel username**
 - you can also use **-r** option to **remove the user's home directory** and **mail spool**

Linux Service/Server Administration

- **Linux Server/Service Administration:** Involves managing and maintaining various services and servers running on Linux-based operating systems
- **Some of the key areas:**
- **System administration:** involves the setup configuration and maintenance of Linux-based systems
- **Responsibilities of Linux System Administrator**
 - **Installing** and **configuring** the operating system, managing user accounts, configuring network interfaces, and managing system resources
 - **File and Print Services** : involve sharing of files and printers across a network
 - Provide file and print services using various protocols, such as **NFS** and **Samba**

Linux Service/Server Administration ...

- **Web Services:** involve **hosting** and **serving web content**, such as **web pages**, **web applications** using various web servers e.g. **Apache server**
- **Database Services:** **storing** and **managing** data using a **database management system (DBMS)** e.g **MySQL** and **PostgreSQL**.
- **Mail Services:** **sending** and **receiving email messages** using mail servers, such as **Postfix** and **Exim**
- **Virtualization:** running multiple virtual machines on a single physical machine e.g. **KVM** and **Xen**.
- **Security and monitoring:** protecting systems and data from unauthorized access and monitoring system performance and health e.g. **firewalls** and **intrusion detection systems (IDS)**.

Supporting a Windows Network - through SAMBA

- **Samba** is a free and open-source software allows Linux and Unix systems to *share files and printers* with windows computers .
- Provides file and print services to windows clients using SMB/CIFS protocol.
- **System Administrators** use Samba to **integrate Linux/Unix systems with Windows networks, manage access controls and permissions.**
- To configure Samba for file and printer sharing, the administrator needs:
 - *Install Samba on the Linux/Unix system*
 - *Configure Samba to share directories and printers*
 - *Set up user authentication and access controls*

Supporting a Windows Network - through SAMBA...

- **Samba security considerations**

- *Samba has several security features to restrict access to shared resources*

These include:

- **User authentication:** *use several authentication methods, including local user accounts, Active Directory, and LDAP.*
- **Access Control:** *use access control lists (ACLs) to restrict access to shared resources based on user/group membership and permissions.*
- **Encryption:** *use encryption to secure data transmissions between the Linux/Unix system and Windows clients.*

Mail Server

- *Email systems consist of **mail servers** and **mail clients**.*
- ***Mail servers** are responsible for **storing** and **forwarding mail messages***
- ***Mail clients** are used to **read** and **compose email messages**.*
- *To set up and configure a mail server, administrator needs to:*
 - *Install a **MTA** (Mail Transfer Agent) to handle incoming and outgoing mail e.g. Postfix or Sendmail*
 - *Configure **MTA** to handle mail delivery, including DNS configuration and mail routing*
 - *Install a **MDA** (Mail Delivery Agent), to handle mail storage and retrieval e.g. Dovecot or Courier*
 - *Configure the **MDA** to handle user authentication and access controls.*

End of Chapter 3

Thank You!

Any Question ?